

Source Code

#LAB 3.4 Regular Expression for Identifying the Structure of Input

(a+b)*aa(a+b)*

```
import re
```

```
def identify(str):
```

```
    pattern=r"^[ab]*aa[ab]*$"
```

```
    return re.fullmatch(pattern,str)
```

```
str=input("Enter a string to test: ")
```

```
if identify(str):
```

```
    print("Accepted")
```

```
else:
```

```
    print("Rejected")
```

Output

```
Enter a string to test: abababaaabababab
Accepted
○ (base) PS D:\Prasoon Raj Shakya\BSc. CSIT\4th sem\Lab works>
```

```
Enter a string to test: ababababa
Rejected
○ (base) PS D:\Prasoon Raj Shakya\BSc. CSIT\4th sem\Lab works>
```

Conclusion

The program successfully implements the regular expression $(a+b)^*aa(a+b)^*$ over the alphabet $\{a,b\}$. It accepts strings containing aa and rejects strings that do not satisfy the required structure.